

IN COUNTERFACTUALS WE'RE ALL DEAD

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"What happened, happened, and couldn't have happened any other way"

ABSTRACT: In his important contribution, Jörg Guido Hülsmann (2003) attempted to redefine the method of economic science. According to him most economic laws are counterfactual in their nature and do not have to rely on *ceteris paribus* assumptions. The purpose of this note is to demonstrate the shortcomings of counterfactuals, and to reveal that Hülsmann's contribution is based on the broad *ceteris paribus* concept. In the first section, we shortly describe the *ceteris paribus* rule. In the second, we approach the problem of counterfactuals. Third section introduces the *ceteris paribus* elements into the counterfactual analysis. The article ends with conclusions.

KEYWORDS: counterfactuals, methodology

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INTRODUCTION

Jörg Guido Hülsmann (2003) proposed to substitute the old “*ceteris paribus*” clause for what appears to be a broader “than otherwise would have been” clause. His argument appears to be that the *ceteris paribus* rule narrows economic laws to specific circumstances in which almost all factors remain constant. The counterfactual modification, then, allows us to increase the range of applications for economic laws, since it allows other discussed economic factors to change in theoretical considerations. His conclusions, however, depend on the understanding of notions specific for each approach. In other words, the meaning that lies beneath “than otherwise would have been” and “all other things equal” resolves the supposed conflict between the two. As we plan to show, counterfactual statements always need some additional assumptions, which make the proper counterfactual approach look very similar to *ceteris paribus* framework. The *ceteris paribus* clause on the other hand also has two possible different versions; the narrower one, based on strict “equilibrium” states (such as an evenly rotating economy); and the broader one that can be reconciled with the proper counterfactual theme.

1. *CETERIS PARIBUS* SETS THE STAGE

Causality is the cornerstone of scientific reasoning. Economic science delivers explanations and reasons why selected causes imply the existence of necessary (or probable) results. The main problem is to separate those “essential” connections from “accidental” ones, especially in the environment where many factors come into play. In chemistry during the investigation of an oxygen and hydrogen mixture, different experimental amounts might be mixed under a constant temperature. Temperature shifts may lead to various additional results—instead of synthesizing water, one can cause an explosion. In order to avoid unpredictable differences in results—that is in order to put aside *other* aspects for a different analysis—the scientist has to keep *other* factors constant, so that they do not interfere with the main experiment. This is the reason for usage of one of the most important scientific assumptions, *ceteris paribus*, or all other things equal.

Notice that in a typical chemical experiment this “equal” usually means “the same,” or “unchanged,” so that the quantity and quality of other factors remains intact. In the example of the oxygen and hydrogen mixture, the temperature is the same. It also has to be the case with the equipment, and all the things that may be relevant for the final result. In other words, *ceteris paribus* is very strict and narrow. We start from an equilibrium point, where everything appears to be unchanged, and then we can examine the consequences of one introduced change.

In economics, at least on the surface, matters seem similar. Since there are many different factors affecting the economic state of affairs, one needs an assumption about how to separate the object of analysis, and rule out redundant things.¹ Let us take the example of a theorem demonstrating the effects of taxation on capital accumulation. The phenomenon of capital accumulation is not dependent only on one factor. It is an outcome of many individual decisions, which are themselves influenced by a wide range of factors. In the theorem a set of other factors has to be kept “constant” so that only the changing factor will cause a change in the amount of accumulated capital.²

Going further in this hypothetical case, the economist postulates an “economic law”—an increase in taxation leads to a decrease in capital accumulation, *ceteris paribus*. Suppose that we describe capital accumulation as in the following equation:

$$K = f(A, B, C, D, T)$$

¹ The fact that chemistry is based on natural experiments and that economics might be considered to be based on the so-called “thought experiments,” is for the moment beside the point. The justification of *ceteris paribus* assumption works in both cases similarly. In our ending section, however, we reach a conclusion that these are radically different.

² The interesting thing relevant for the current discussion is that, as Persky shows, the *ceteris paribus* concept just filtered into economic treatises and economic language without a deep systematic explanation of what it really means to “keep things the same” (Persky, 1990). Lack of systematic exposition explains misunderstandings about the concept. In *Man, Economy, and State* Rothbard (2004) uses “*ceteris paribus*” 32 times. In *Human Action* Mises (1966) uses “other things being equal” 27 times. Yet they do it without systematic explanation and exposition. Despite Mises and Rothbard’s rigorous analyses on other points, they fail to carefully define those concepts and decide to treat them as obvious.

Letters A, B, C, D, T denote different factors affecting the amount of accumulated capital (K).³ Factor T represents the influence of taxation. The narrow *ceteris paribus* approach assumes that everything except T has to remain unchanged so that all other things do not influence or induce a change in K (because they stay the same, any change in K cannot be attributed to them). The one and only factor changed is taxation, hence every change in K will be attributable solely to change in T. Breaking the *ceteris paribus* postulate would cause problems for such strict theoretical statements. If other factors are allowed to change, say A and B go up while C and D go down, then we do not have the means to determine the effect of K on T. We cannot be sure whether the resulting change of K was caused either by a change in A, B, C, D, or T, or a combination of those. *Ceteris paribus* resolves this issue and allows for formulation of meaningful economic laws in relationships like these.

In the classical approach the law is depicted as follows: *an increase of taxation leads to a decrease in the amount of private capital ceteris paribus*, or *an increase of taxation leads to a decrease in the amount of private capital all other things equal*. The equation for this would be:

A, B, C, D (other factors) = constant

$K \downarrow = f(A, B, C, D, T \uparrow)$

Provided that all the factors except taxation stay the same through the “experiment,” a decrease in the amount of private capital has to be attributed to the only factor that has changed, i.e., taxation.

It is easy to see a problem with the narrow *ceteris paribus* approach. *Scientific laws describing functional or causal relationships under specified conditions refer to particular cases in reality only when those conditions are met in reality* (unless something more is specified). If the conditions of the acknowledged law are not met in reality, then the law is not directly applicable. As in chemistry, if the interaction between oxygen and hydrogen is said to appear under a certain constant temperature, then when in the real case the temperature is constantly changing, the law cannot explain what is happening

³ The equation used here is only for pedagogical purposes. We do not argue that capital can be measured. We plan to illustrate the connections rather than arrive at mathematically operational model of capital. It will help us to grasp the essence of Hülsmann’s point.

to oxygen and hydrogen. The case with economics is similar—the law “if the taxes are increased and *all other conditions stay the same*, then the amount of capital will be lower,” refers to reality only if the stated conditions (in italics) are met in reality, i.e., if nothing else is changed. Since it can hardly ever be the case that all conditions stay the same (A, B, C, D, other factors, stay intact), it appears that the law has a very small range of application. In other words, once other conditions change, the law *in this particular form* does not tell us about the influence of increased taxation on the level of accumulated capital.⁴

2. COUNTERFACTUALS AND THEIR LIMITS

We can see the advantages of Hülsmann's point.⁵ Economic laws can be stated in a counterfactual manner—we do not need to assume that other conditions should stay the same for the casual connection to work. In order to restate the above law and move away from *ceteris paribus* clause, we could say “increased taxation leads to a lower amount of accumulated capital *than otherwise would have been the case*.” Even if other things change, and also influence capital accumulation, we can still grasp the damaging effect of taxation upon it. Suppose the higher propensity to save increased accumulated private capital, and at the same time higher taxation was introduced. Even though after those two changes the amount of saved capital might be higher, we still know that higher taxation led to lower capital accumulation than otherwise would have been.

The importance of Hülsmann's contribution seems obvious and intuitive—most economic laws we know could refer not only to conditions when all other factors remain the same. Hence, those

⁴ Sometimes an even more extreme assumption than *ceteris paribus* is used in economics: *ceteris absentibus*, where some factors are not held constant, but are completely removed from the picture (Joseph, 1980, p. 777; Pietroski and Rey, 1995, p. 104). This is the case of general equilibrium theory (or the Austrian “Evenly Rotating Economy”)—its application to the real world is a puzzle. Long (2005) following Aristotle calls *ceteris absentibus* “precise abstractions.”

⁵ There is a vast body of philosophical and logical literature on subjunctive conditionals and counterfactuals. Two publications are especially worth mentioning: an overview of the literature since World War II (Collins et al., 2004), and Bennett's guide (2003).

laws are applied to wider range of cases than a narrow *ceteris paribus* assumption would seem to allow.⁶

One of the shortcomings of the counterfactual approach—that Hülsmann did not consider in his article—is what exactly is hidden behind the phrase “otherwise would have been the case.” Which *alternative* scenario are we analyzing? Since we drop the “all other things constant” rule, there are many other possible changes we can speculate about. How do we distinguish among those changes? To put it differently, *to every fact we could try to invent more than one counterfactual scenario*. We are left with the problem then, which other possible world that might have existed is relevant to our theoretical considerations? There are numerous *possible worlds*⁷ in the counterfactual ladder—which possible world does Hülsmann advise us to hide behind the phrase “otherwise would have been”? Certainly he cannot have in mind the whole set of all the possible worlds that could have existed.

To try to answer this question, let us go back to the equation illustrating the casual mechanism of capital accumulation. Consider the case in which, apart from increased taxation, other conditions are changed—factors A, C and D go up, whereas B falls. The modified equation would be the following:

$$K = f(A\uparrow, B\downarrow, C\uparrow, D\uparrow, T\uparrow)$$

In this dynamic setting we have four other factors changing. A counterfactual statement on tax influence suggests that K is lower than in *the case in which tax was not increased*. Let us here consider the case in which taxes were not raised, and label the amount of capital in this scenario as K':

⁶ Another thing is that in most economic writings, authors intuitively use a counterfactual approach, and do not really stay faithful to the hard *ceteris paribus* rule.

⁷ On the problem of possible worlds see Sanford (1989, p. 101ff); Lewis (1973, p. 559). For one of many examples of how problematic the notion of similarity of worlds is, see Hausman (1996). Quine eloquently expressed his reservations about the notion of counterfactuals: “Take, for instance, the possible fat man in that doorway; and, again, the possible bald man in that doorway. Are they the same possible man, or two possible men? How do we decide? How many possible men are there in that doorway? Are there more possible thin ones than fat ones? How many of them are alike? Or would their being alike make them one? Are no two possible things alike? Or finally, is the concept of identity simply inapplicable to unactualized possibles?” (Quine, 1953, p. 4).

$T = \text{constant}$

$K' = f(A?, B?, C?, D?, T)$

The counterfactual framework would tell us that $K' > K$, or that without higher taxes, more private capital is accumulated. The real problem, however, is that the correctness of this counterfactual sentence depends on what is hidden behind the question marks, or how other factors behave in the alternative scenario. Depending on how we fill in those blanks, we create a possible world that might have been realized. The appropriateness of our law depends on which world is envisioned. The suggested answer would be that in the alternate scenario other factors have to change in the same way as in the actualized scenario of increased taxation:

$K' = F(A\uparrow, B\downarrow, C\uparrow, D\uparrow, T)$

Under these assumptions, the counterfactual statement will be correct. K' is higher than K , because of a negative impact of taxation upon the amount of capital. The point is that in the alternate scenario other factors, i.e., A , B , C , D , have to change in the same way as in the realized scenario. Factors do not have to *stay the same*, but in counterfactual scenarios they have to *change in the same way as in the factual scenario*. That is why the counterfactual approach can be seen as a broader *ceteris paribus* assumption. It allows other conditions to change. Nevertheless, they cannot change randomly from scenario to scenario. Rather, they must be in some way fixed, or controlled.⁸ In other words, counterfactuals should and could use the notion of *ceteris paribus*, but it would have to be called all other things *considered*, not all other things *unchanged*.

In their criticism of Hülsmann's paper, Stringham and Gonzales (2009, p. 9) argue that his approach, compared to *ceteris paribus*, is "less accurate because it basically attempts to disregard changes in all other variables." Hülsmann exposed himself to such criticism, when he specified counterfactual laws "as absolutes that are not conditioned by any other factors" (2003, p. 74). The solution seems to be in the middle ground. As we saw above, a counterfactual has a certain advantage over *ceteris paribus*, because it allows other

⁸ A softer, less static, version of *ceteris paribus* used in alternative scenarios therefore abstracts from "disturbing causes" that might be developed in the counterfactual.

variables to change. Nevertheless, it is not possible to go all the way and disregard all the other variables. They certainly have to be contemplated, yet they do not have to remain completely unaltered as in the extreme *ceteris paribus* clause.

Moreover, in many cases several variables depend on each other and interact, inducing other changes, which also induce other changes, etc.⁹ The above example of taxation is an easy case of independent variables. Cases of interdependent variables require more scrutiny and careful analysis.¹⁰

3. *CETERIS PARIBUS* STRIKES BACK?

It appears that in counterfactual “thought experiments,” the causal chain between the variables has to be “controlled.” Counterfactual statements rely on the broader *ceteris paribus* clause—some version of *ceteris paribus* limitation has to be placed on other variables in alternative scenarios. One has to envision the “otherwise” case, a contrary world to what has actually happened. Which particular possible world should we choose in order to demonstrate a valid counterfactual? The *ceteris paribus* approach comes to the rescue. As one of the commentators observed:

But nothing in the notion of similarity *per se* tells us what it means to be similar enough, hence close enough. To what, then, do we appeal? I suggest we use *ceteris paribus* concept. I translate this Latin phrase as “other things being equal,” and aver that it yields more insight about

⁹ Therefore, while dealing with conditionals one has to avoid the trap of a “conditional fallacy,” which is explained by Shope as a case in which “...one has overlooked the fact that, in some of the specified situations, statement *p* is actually true but if *a* were to occur then the occurrence of *a* would be at least part of what would make *b* absent (make *b* occur) or *a* would occur at least partly because of the absence of *b* (occurrence of *b*) or because of what makes *b* absent (makes *b* occur) or would together with absence of *b* (occurrence of *b*) form at least part of reason for some other occurrence” (Shope, 1978, p. 403).

¹⁰ One could argue that the case of interdependent variables poses a threat to the counterfactual approach. But this is not really a threat, since it is a challenge for any theoretical approach, be it counterfactual in nature or placed within a *ceteris paribus* scheme. If in the real world we know that some variables are interdependent, then if we use *ceteris paribus* clauses we simply assume that problem away and refrain from investigating it. This is hardly a solution to the interdependence puzzle.

what makes a counterfactual true or false than similarity, closeness, degree of change.... (Mikkelsen, 1996, p. 196)¹¹

To make the taxation theorem relevant for counterfactual scenarios, we have to say that in addition to abstracting from *historical* disturbances, we also have to keep *other theoretical disturbances* constant. Hence the counterfactual statement on taxation could be the following: "increased taxation leads to lower amount of private capital than otherwise would have been the case, provided that in the alternative scenarios all the factors relevant for the casual process involved would have *changed in the same way* as in the actual scenario." Compare this with the previous hard *ceteris paribus* rule that could safely state, "increased taxation leads to a decreased amount of private capital, provided that all the factors relevant for the casual process involved would have *remained the same* all the time and, under those stable conditions, would not endogenously cause any change."¹²

The soft *ceteris paribus* principle, the assumption about the movement of controlled variables in alternate scenarios (a sort of *mutatis mutandis*), is indispensable for the counterfactual framework to work. Otherwise, counterfactuals are nothing other than historical speculations.

CONCLUSION

In the light of the preceding sections, one contribution of Hülsmann's paper seems unquestionable—his point that many economic variables described in economic laws do have a recognizable impact upon economic reality even if other variables are changing (although not in all conceivable cases, of course).¹³ A

¹¹ Robert Nozick proposed another solution. His proposition was to consider not only a closest possible world, but the closest possible bundle of worlds. See Nozick (1981, p. 680, n. 4). I am grateful to Dr. David Gordon for this point.

¹² We should also remember there are laws in economics which require other variables to indeed stay constant and completely unchanged (Hülsmann, 2003, p. 77, n. 24).

¹³ Overlooking this fact would be, as Shope notes (1978, p. 406), a *ceteris paribus* fallacy since "those factors which would block the occurrence of *b*, were *a* to occur, may themselves have no particular connection with the occurrence or

discussion on this impact in cases that *might have been*, however, is going a step further, requiring a lot more caution.

Our inquiry leads us to additional reflections. *Ceteris paribus* clauses and counterfactual scenarios are not really building blocks of economic science. Building blocks are “natures” of things that we know *prior* to working with *ceteris paribus* and counterfactual examples. We already know the impact of taxation on the amount of capital, before we apply it to *ceteris paribus* cases, and we already know the impact of historical disturbances or the impact of other variables, when we play with different counterfactual scenarios.

There are two types of this prior knowledge in economics. One group of laws is associated only with the action axiom (those laws are on means, ends, scarcity, subjective utility, etc.—already implied by the axiom). The other set of laws requires additional empirical considerations and deals with causal chains (those laws are on the pricing process, nature of money, growth, etc.). It is for the second group of laws that we conduct thought “experiments.” Yet they are not really experiments, because the results are grasped with the aid of internal reflection.¹⁴

The thesis might be proven by showing how the *ceteris paribus* clause in economics radically differs from *ceteris paribus* in natural sciences. In the natural sciences, an experiment consists of two types of actions. At first, the state of affairs is stable and unaltered by any spontaneous or endogenous change (empirical equilibrium is in place) hence all the variables remain “constant.” Then one variable is altered in the course of an experiment and effects follow. The comparison of two states of affairs refers to narrow empirical observation. It is the empirical observation that is the standard for scientific judgment. The experiment in natural sciences is undertaken without full prior knowledge. First we observe an empirical equilibrium, and then one change in the variable starts off a chain of events.

In economics, on the other hand, with regard to the *ceteris paribus* clause, there are no narrow observations, but two explorations.¹⁵

absence of *a*. That is, it may just be a coincidence that they would be present if *a* were to occur.”

¹⁴ Thanks are due to an anonymous referee for this point.

¹⁵ This statement applies to those who believe that economics is a science prior to history (Austrians).

The first is that a change in one variable leads to recognizable consequences if other factors remain constant. But there is also a second implicit exploration, that if all things remained constant with no changes in variables, equilibrium forces would prevail. In other words, in *ceteris paribus* economic laws there is an implicitly assumed thesis that the economic system with the existing set of unchanging variables will not cause any change *endogenously*. Contrary to natural sciences, where the starting point is the uncertainty of outcomes, *ceteris paribus* laws in economics begin with this *prior knowledge*, because there is no reference to direct empirical observation, while conducting a thought experiment. This demonstrates that connections between entities in economics are not recognized in *ceteris paribus* vacuums (or counterfactuals), because we apply previously recognized knowledge of those to *ceteris paribus* examples to *illustrate* them. Or to put it differently, the so-called "thought experiments" in economics are not experiments at all, because in *ceteris paribus* cases we refer not to narrow empirical observations, but instead to prior knowledge of causes and consequences inferred from natures of things.¹⁶

To conclude, Hülsmann made an important addition to economic method. Most laws that economists usually present under the *ceteris paribus* assumptions can be presented in a broader framework of counterfactuals. Not all types of counterfactuals, since it is crucial to distinguish between various possible scenarios. Sometimes, changes of other factors do not overcome the influence of the analyzed factor. The *ceteris paribus* assumption and counterfactuals are not poles apart. *Ceteris paribus* itself needs a counterfactual approach, and counterfactuals need some soft version of the *ceteris paribus* assumption (or otherwise they lead to a kaleidic and accidental universe).

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¹⁶ "...any counterfactual about a particular event implies or presupposes something about the event's essence" (Bennett, 1987, p. 369). The same is to be said about *ceteris paribus* not assumed in narrow empirical experiments, but by rational investigations.

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